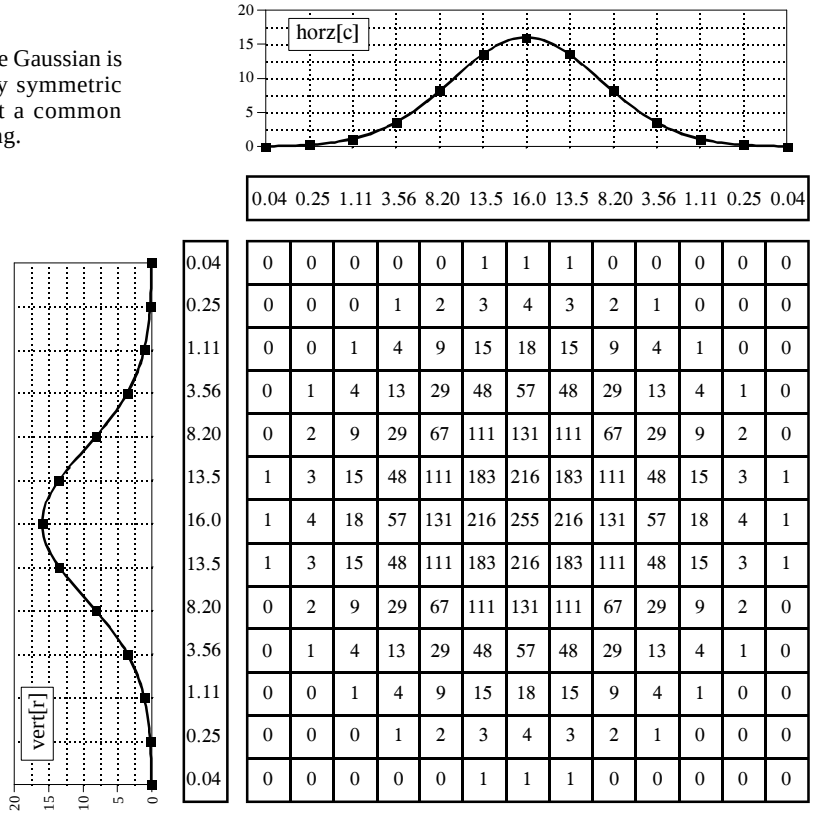

FIGURE 24-7
Separation of the Gaussian. The Gaussian is the only PSF that is circularly symmetric *and* separable. This makes it a common filter kernel in image processing.

but there is only one, the *Gaussian*. As is shown in Fig. 24-7, a two-dimensional Gaussian image has projections that are also Gaussians. The image and projection Gaussians have the *same* standard deviation.

To convolve an image with a separable filter kernel, convolve each *row* in the image with the *horizontal projection*, resulting in an intermediate image. Next, convolve each *column* of this intermediate image with the *vertical projection* of the PSF. The resulting image is identical to the direct convolution of the original image and the filter kernel. If you like, convolve the columns first and then the rows; the result is the same.

The convolution of an $N \times N$ image with an $M \times M$ filter kernel requires a time proportional to N^2M^2. In other words, each pixel in the output image depends on *all* the pixels in the filter kernel. In comparison, convolution by separability only requires a time proportional to N^2M. For filter kernels that are hundreds of pixels wide, this technique will reduce the execution time by a factor of *hundreds*.

Things can get even better. If you are willing to use a *rectangular* PSF (Fig. 24-5) or a *double-sided exponential* PSF (Fig. 24-6), the calculations are even more efficient. This is because the one-dimensional convolutions are the *moving average* filter (Chapter 15) and the *bidirectional single pole* filter